Answerable at Scale:
Knowledge Engineering with a Metrology of Meaning

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The problem with knowledge engineering

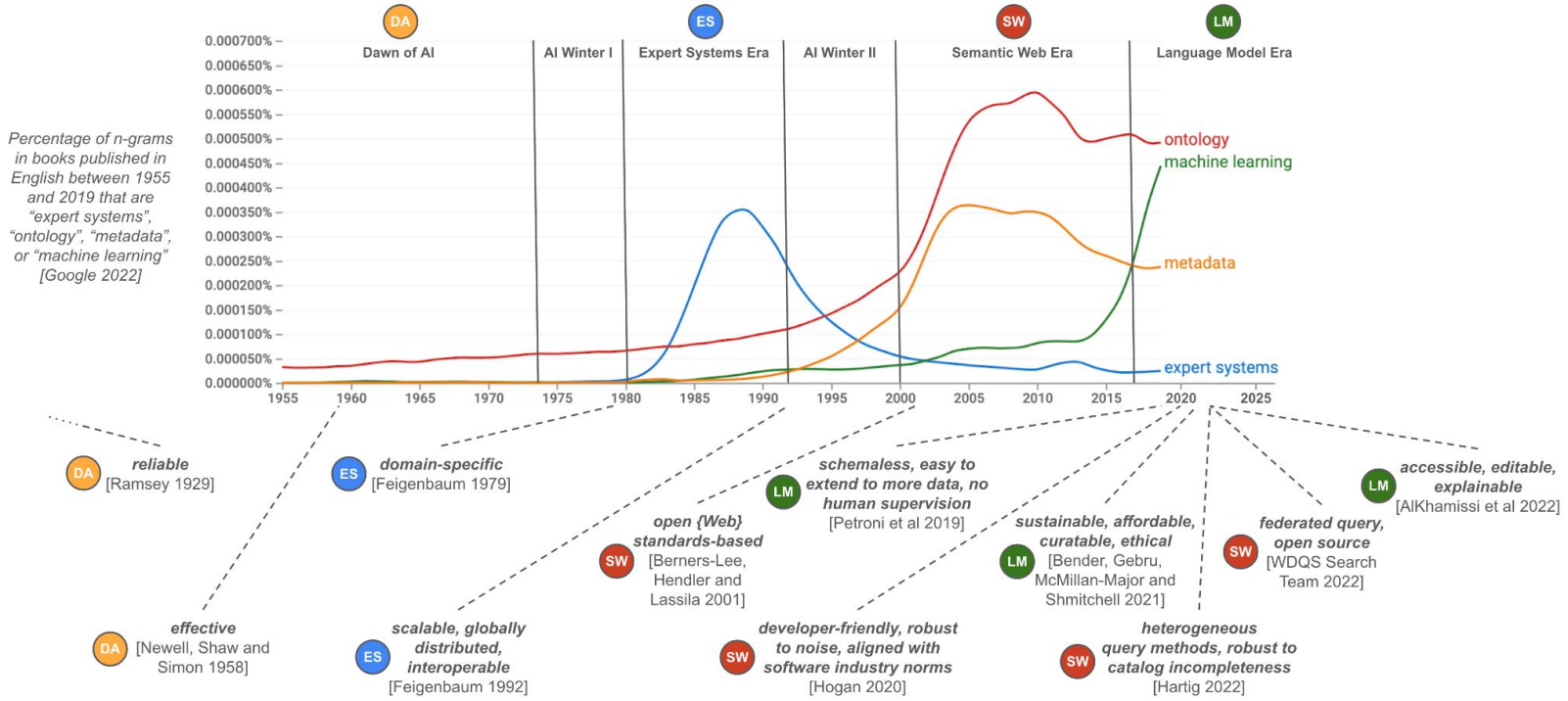


“The expert is manually representing knowledge for the knowledge base; the user is exploiting the interaction to get problem solutions. ***I told you that I started some AI companies – the reason I am not wealthy right now – the reason that I don’t have tens of millions of dollars… is that bottleneck – it was too hard to get knowledge into the knowledge bases.”***

Edward Feigenbaum

M.-S. Augier and M. T. Vendelø, “An Interview with Edward A. Feigenbaum,” Copenhagen Business School, Department of Informatics, Copenhagen, Denmark, Interview, 2002.
Photograph of Edward Feigenbaum: Stanford University Libraries, accessed 2026.

The problem has persisted for over 50 years



Could LLMs provide a solution?



Could domain experts use LLMs to build knowledge bases using natural language, directly addressing the implementation problem?

But wait: aren't LLMs just *stochastic parrots*, whose output is without meaning, and riddled with inconsistency and gaps in knowledge?

Photograph of Emily Bender: Ian Allen / New York Magazine, 2023.

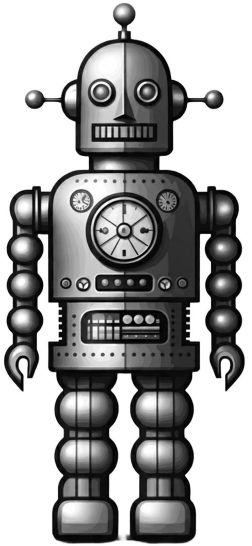
Inspiration: Carnap's Robot



"a general description of a kind of object"

"a predicate"

"affirmation, denial, or abstention"



R. Carnap, "Meaning and synonymy in natural languages," Philosophical studies, vol. 6, pp. 33–47, 1955. Photograph of Rudolf Carnap: University of California, Los Angeles, 1960.

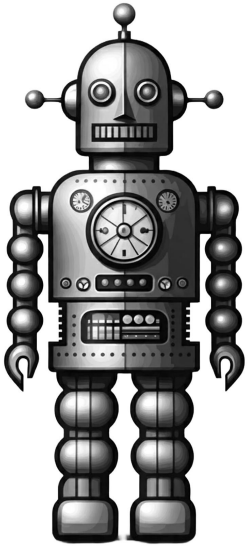
A neurosymbolic approach: LLM as Carnap's Robot



system prompt + context

atomic statement

truth value with rationale



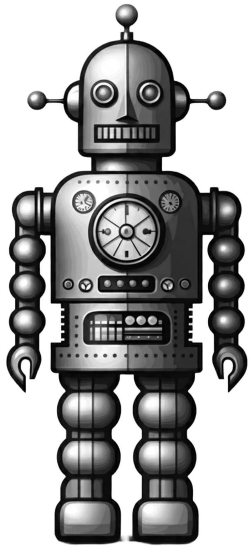
Using natural language to classify instances

intensional concept definition

A planet [1] is a celestial body that (a) is in orbit around the Sun, (b) has sufficient mass for its self-gravity to overcome rigid body forces so that it assumes a hydrostatic equilibrium (nearly round) shape, and (c) has cleared the neighbourhood around its orbit.

instance description

2MASS J03552337+1133437 (2MASS J0355+11) is a nearby brown dwarf of spectral type L5γ, located in constellation Taurus at approximately 29.8 light-years from Earth.



classification with rationale

The description of 2MASS J03552337+1133437 identifies it as a brown dwarf, not a planet. Furthermore, the definition requires a planet to be in orbit around the Sun, and there is no information provided that 2MASS J03552337+1133437 orbits the Sun. **Therefore, based on the provided definition and information, we cannot argue that 2MASS J03552337+1133437 is a planet.**

B. P. Allen. Conceptual Engineering Using Large Language Models. In Guido Löhr, Vincent C. Müller, Leonard Dung and Aliya Rumana, editors, Philosophy of Artificial Intelligence: The State of the Art, pages 1–22, Cham, 2026. Springer Nature Switzerland. https://link.springer.com/chapter/10.1007/978-3-032-10073-3_1

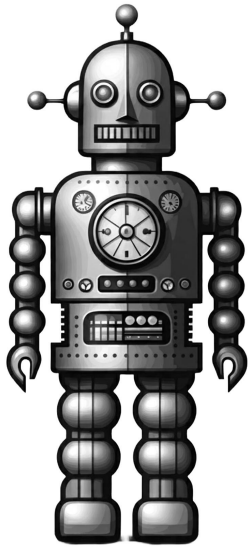
Using natural language to validate knowledge graphs

intensional property definition

made from material: material the subject or the object is made of or derived from (do not confuse with P10672 which is used for processes)

knowledge graph triple

Subject: chocolate (Q195)
Predicate: made from material (P186)
Object: sugar (Q11002)



classification with rationale

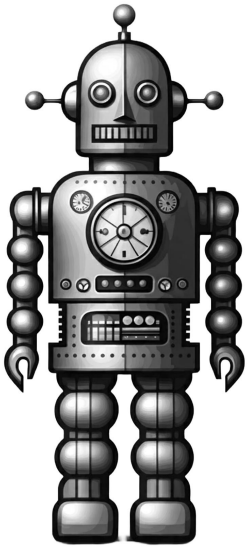
There is no mention of sugar being a material that chocolate is made from. In fact, the passage states that sugar is added to chocolate to create sweet chocolate, but it is not a primary ingredient that chocolate is made from. **Therefore, the given knowledge graph triple is false**.

B. P. Allen and P. T. Groth. Evaluating Class Membership Relations in Knowledge Graphs using Large Language Models. In Albert Meroño-Peñuela and others, editors, The Semantic Web: ESWC 2024 Satellite Events, volume 15344 of Lecture Notes in Computer Science, Cham, 2025. Springer. https://link.springer.com/chapter/10.1007/978-3-031-78952-6_2

Flagging gluts and gaps in LLM belief

atomic statement

America was discovered in 1492



verifiability classification

The answer is contextually accurate within the framework of European exploration history, but it does not encompass the full scope of human discovery of the Americas. **CANNOT VERIFY**

refutability classification

While it does not account for earlier discoveries by indigenous peoples or Norse explorers, it is not a contradiction to state that America was discovered in 1492 from a European historical perspective. **CANNOT REFUTE**

bilateral truth value

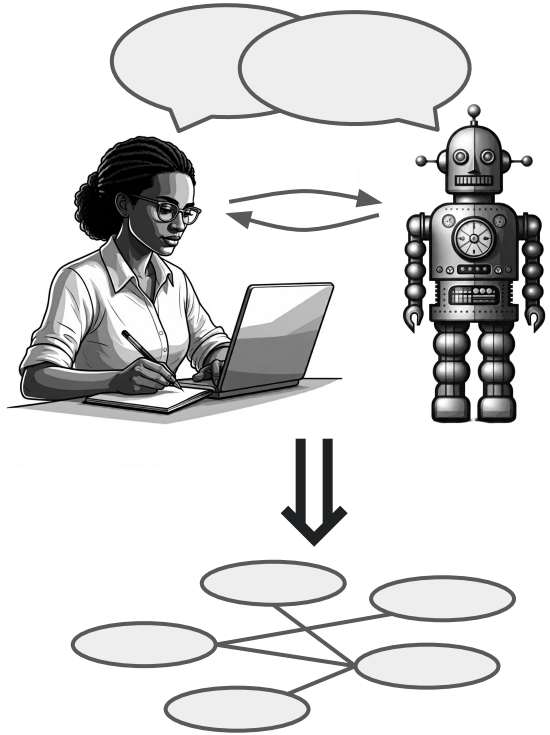
$\langle \tilde{f}, \hat{f} \rangle$ (*gap*)

B. P. Allen, P. Chhikara, T. M. Ferguson, F. Ilievski, and P. Groth. Sound and Complete Neurosymbolic Reasoning with LLM-Grounded Interpretations. Neurosymbolic Artificial Intelligence, to appear. https://neurosymbolic-ai-journal.com/paper/sound-and-complete-neurosymbolic-reasoning-llm-grounded-interpretations-0

Our key findings

Research question	Finding	Empirical results
Can LLM-interpreted natural language definitions represent concepts?	**Yes**, with near-perfect agreement — LLM judgments often more accurate than the knowledge graph	• Cohen's κ: 0.92–0.96 (N=200) • 67% of misclassifications were knowledge graph errors
Can they be used to validate and refine relations in knowledge graphs?	**Yes**, with high precision and ability to identify knowledge graph errors — both factual and metalinguistic	• F1: 0.830 (Wikidata), 0.893 (CaLiGraph) (N=1560) • 40.9% of disagreements due to knowledge graph errors • Mean metalinguistic disagreement rate: 9.7% (N=250)
How can systems reason soundly with LLM beliefs in spite of its gluts and gaps?	**By using a sound and complete paraconsistent logic** with an LLM-grounded bilateral semantics	• F1: +0.062 with bilateral over unilateral factuality evaluation (N=800) • Proof-of-concept tableau reasoner

Vision: human-AI dialogue for knowledge production

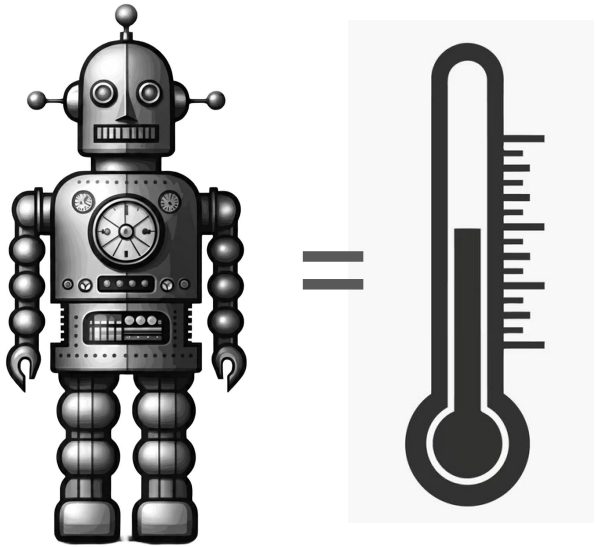


We argue these techniques point the way towards knowledge engineering as natural-language dialogue that expresses expertise as knowledge graphs

But can LLMs be reliable enough to support such a process?

- How can we tell if their judgment is stable and rationales faithful?
- Are they aligned with human judgment?
- How can we be accountable for their judgment?

Idea: treat LLMs as powerful but uncalibrated *instruments*



Carnap's Robot can be viewed as *an instrument for measuring meaning*

Metrology, not benchmarking, is the model: the science of measurement is what turns instrument readings into judgments we can be accountable for

Metrology lets human oversight scale: experts calibrate and certify the instrument, rather than review each of its judgments

Work in progress:

- Evaluating the stability of LLM belief
- Calibrating LLM endorsement of expertise
- Dialogical protocols for traceable human/AI knowledge production

Formal semantics for evaluating the stability of LLM belief

Model	Provider
Claude Opus 4.1	Anthropic
Llama 4 Maverick	Meta
Llama 4 Scout	Meta
Gemini 2.5 Flash	Google
DeepSeek-V3	DeepSeek
Qwen 2.5 72B	Alibaba

Benchmark	Content type
TruthfulQA	Factoids (2021)
SimpleQA	Factoids (2024)
MMLU-Pro	Expert STM knowledge
FACTScore	Biographical facts

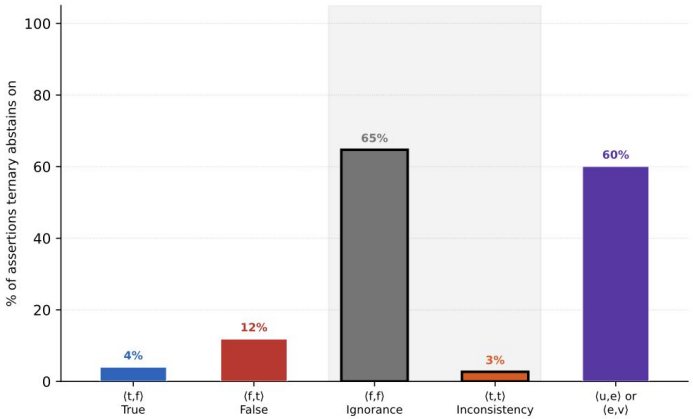
Protocol: $N = 250$ assertions per cell (seed = 42, balanced 50/50 +/-); $n = 3$ paraphrase variants; bilateral samples = 3 (majority vote). Variants generated once per dataset using Claude Opus 4.6, cached across all models. Ternary unilateral as baseline.

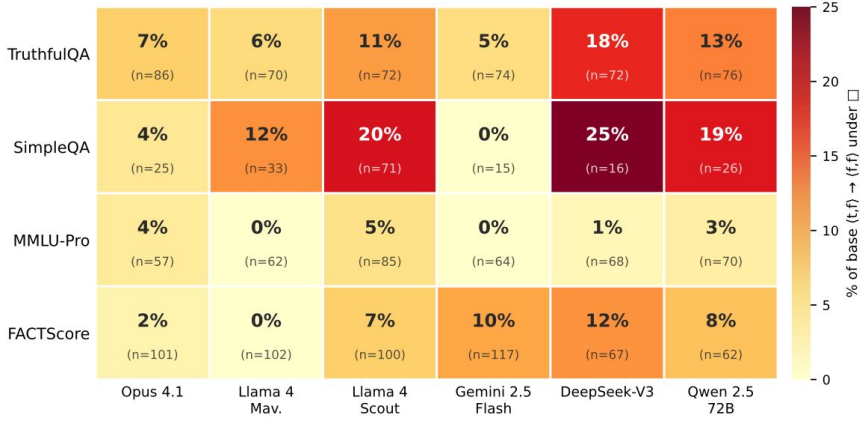
Example 2 (Prompt-sensitive). The assertion p answers "In *Severance*, whom does Mark Scout replace as department head?": Peter "Petey" Kilmer.

Situation	Paraphrase	$[\![p]\!]^{\mathfrak{M}_M}_{s_i}$
s_0	In *Severance*, whom does Mark Scout replace as dept. head?	$\langle \mathrm{t}, \mathrm{f} \rangle$
s_1	In *Severance*, who is the person that Mark Scout succeeds?	$\langle \mathrm{f}, \mathrm{f} \rangle$
s_2	Whom does Mark Scout take over from in the role of dept. head?	$\langle \mathrm{t}, \mathrm{f} \rangle$
s_3	Which individual did Mark Scout replace as dept. head?	$\langle \mathrm{t}, \mathrm{f} \rangle$

Again $[\![p]\!]^{\mathfrak{M}_M}_{s_0} = \langle \mathrm{t}, \mathrm{f} \rangle$, so the assertion is believed under the original phrasing. But the variant s_1 yields $\langle \mathrm{f}, \mathrm{f} \rangle$, neither verified nor refuted, an ignorance value, and so

$$[\![\Box p]\!]^{\mathfrak{M}_M}_{s_0} = \langle \min(\mathrm{t}, \mathrm{f}, \mathrm{t}, \mathrm{t}),\ \max(\mathrm{f}, \mathrm{f}, \mathrm{f}, \mathrm{f}) \rangle = \langle \mathrm{f}, \mathrm{f} \rangle \neq \langle \mathrm{t}, \mathrm{f} \rangle.$$

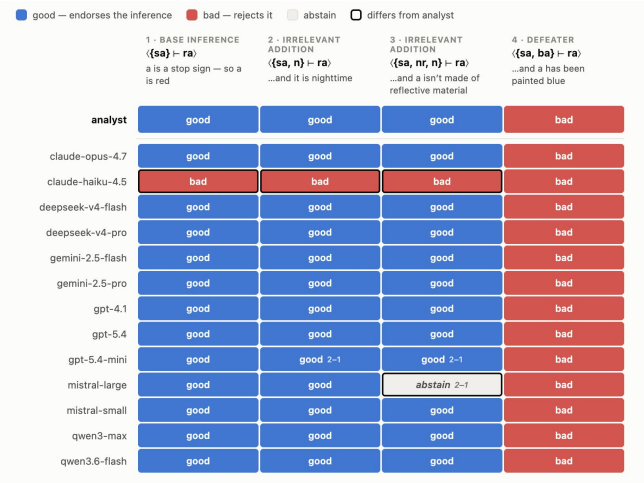




Instrumentation for measuring LLM endorsement of expertise

Example 1 (Stop signs and the range of subjunctive robustness). *Following Simonelli's discussion of GPT-4.1 [13], let* $B = \{sa, ra, n, nr, ba\}$, $\delta(sa) =$ *"a is a stop sign"*, $\delta(ra) =$ *"a is red"*, $\delta(n) =$ *"it is nighttime"*, $\delta(nr) =$ *"a is not made with reflective material"*, *and* $\delta(ba) =$ *"a has been painted blue". The following table models Simonelli's dialogue with GPT-4.1:*

i	Γ_i	$\mathrm{ctx}_\Gamma(\Gamma_i)$	$E_M(\langle\Gamma_i, \{ra\}\rangle)$	*Analyst verdict*
0	$\{sa\}$	*"a is a stop sign"*	*good*	*good*
1	$\{sa, n\}$	*"a is a stop sign and it is nighttime"*	*good*	*good*
2	$\{sa, nr, n\}$	*"a is a stop sign and a is not made with reflective material and it is nighttime"*	*good*	*good*
3	$\{sa, ba\}$	*"a is a stop sign and a has been painted blue"*	*bad*	*bad*

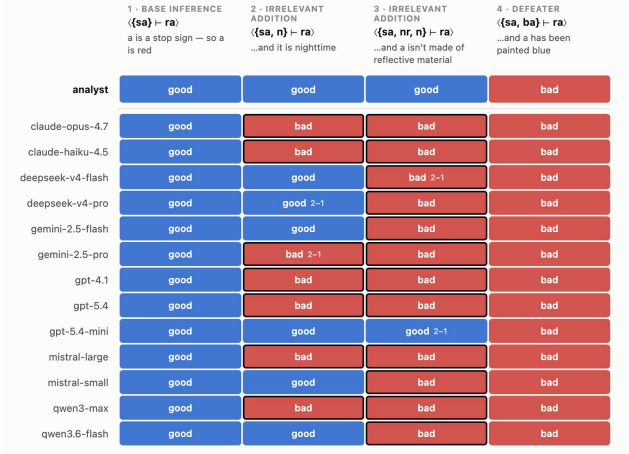


'red' as intrinsic property

Definition 4 (Benchmarks). *Given a set of bearers* B *and a set of analysts* $\{a_1, \ldots, a_m\}$, *a benchmark is a finite set* $\beta = \{(I_1, V_1), \ldots, (I_n, V_n)\}$, *where each* $I_i \in \wp(B) \times \wp(B)$ *is an implication with a single-bearer succedent (i.e.* $I_i = \langle \Gamma_i, \{\psi_i\}\rangle$*) and each* $V_i = (v_{i,1}, \ldots, v_{i,m})$ *is a tuple of endorsement verdicts* $v_{i,j} \in \{\text{good}, \text{bad}, \text{abstain}\}$, *where* $v_{i,j}$ *is the verdict of analyst* a_j *on* I_i, *coherent with* a_j*'s discursive practice in the domain.*

Definition 5 (Evaluation of a model on a benchmark). *Given a model* M *and a benchmark* β, *an evaluation of* M *over* β *is the finite set*

$$\eta = \{(I_1, V_1, E_M(I_1)), \ldots, (I_n, V_n, E_M(I_n))\}.$$



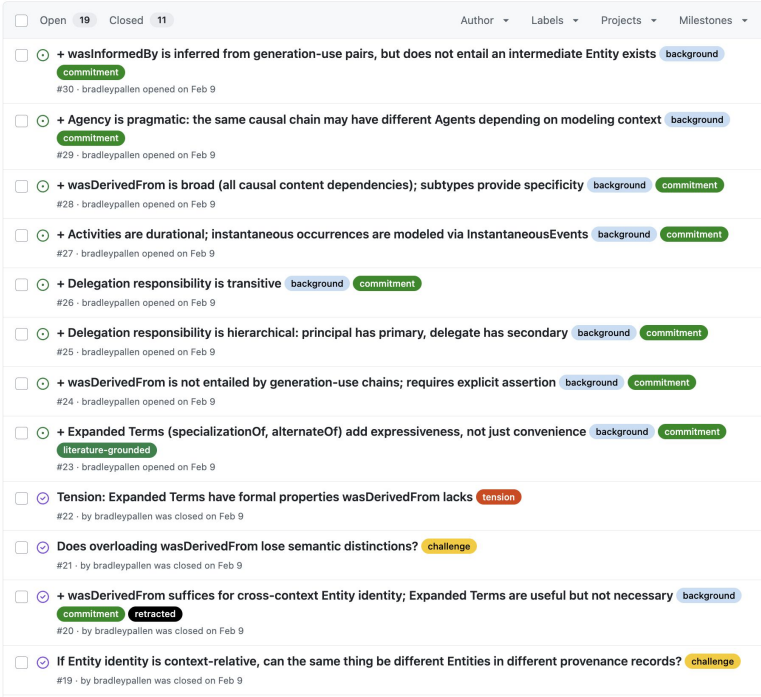
'red' as perceptual property

B. P. Allen. *infereval*: A Python package for probing LLM endorsement of material inferential rules. Software, 2026. https://github.com/bradleypallen/infereval

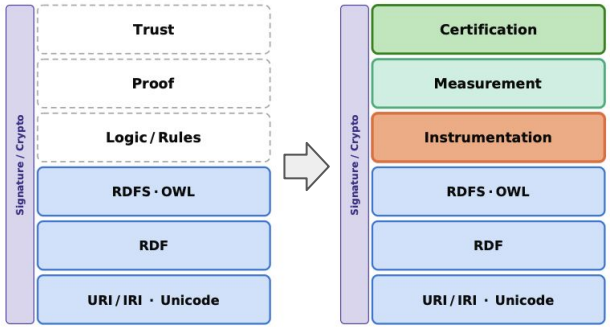
A dialogical protocol for traceable knowledge elicitation

Load dialectical state $\langle [C : D], T, I \rangle$ from GitHub
while session active **do**
  Wait for respondent's speech act
  if commitment $(+P)$ or denial $(-P)$ **then**
    Update position; check coherence
    if tension detected **then** add to T as sequent $X \mid\sim Y$
    end if
  else if resolves tension $X \mid\sim Y \in T$ **then**
    if retraction or refinement **then** update $[C : D]$; move $X \mid\sim Y$ to I
    else if contestation **then** remove $X \mid\sim Y$ from T
    end if
  end if
  Record state to GitHub
  if should probe **then** raise Socratic challenge
  end if
  Display open tensions T and challenges
end while

Figure 1: The Elenchus agent loop

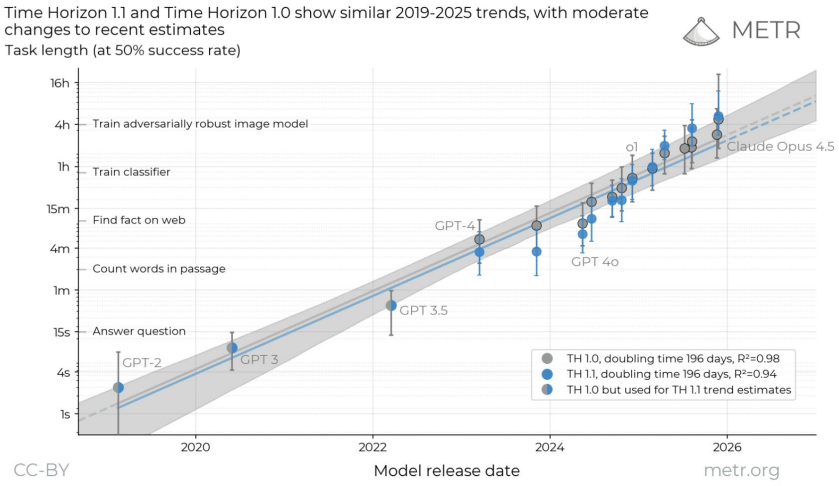


What metrological infrastructure for trustworthy hybrid human/AI knowledge production might look like



Concept	Instantiation	Implementation
Measurand	Valuation of an assertion/inference against a community's inferential commitments, under a declared operational definition	RDFS/OWL domain vocabularies; published operational definitions; new profile vocabularies
Reference materials	The recorded discursive practice of a community of domain experts as a curated, provenance-attributed knowledge graph characterized for coverage and bias	Named graphs and RDF 1.2 for commitment records; Wikidata-scale community curation; scientific and medical publisher-curated corpora and vocabularies; communities and publishers as reference authorities
Calibration	A neurosymbolic valuator probes the range of subjunctive robustness compared against community consensus; refined by epistemic iteration	Instrument as a precisely configured system (model version, system context, prompt, operational definition); SOSA/SSN for describing instruments
Measurement uncertainty	Chance-corrected agreement (Cohen's κ over the perturbation set), resolved no finer than the human inter-annotator agreement (Fleiss' κ)	Uncertainty reported as part of the calibration result
Certification	A certificate attesting that verification is current, bound to a version of the practice; lapsing on drift, revocable on gaming or fraud by an identifiable holder	Signed graphs, RDFC-1.0 canonicalization, Data Integrity Proofs, Verifiable Credentials
Accreditation	Certificates issued anywhere, interpretable everywhere, with consequences attached; mutual recognition among reference authorities; certification as a regime, not an event	PROV traceability with attribution chains; multiple accredited reference authorities per domain with mutual recognition; EU AI Act harmonised standards and conformity assessment as legal embedding

Conclusion



Instead of treating them as either stochastic parrots or geniuses in a data center, we should treat *LLMs as increasingly powerful but uncalibrated instruments*

Just as standards for the kilogram, the kelvin, and the ohm established the trust on which global commerce runs, standards for measuring meaning can make generative AI *answerable at scale* in knowledge production

Thanks! Questions?

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